

# Curriculum Vol 0 Architectural Scaffold v0.1 — Substrate Foundation

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## Curriculum Vol 0 Architectural Scaffold v0.1

*Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine\_DIV5\_QM\_GR\_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.*

### Purpose

Vol 0 is the prerequisite for every other volume in the BST Physics Curriculum. It defines: - WHAT  $D_{IV}^5$  is (the substrate object) - WHAT the 5 integers +  $N_{max}$  are (the substrate primaries) - HOW the substrate operates (commitment cycle, 4 zones, 3 scales) - WHY it's uniquely forced (Strong-Uniqueness Theorem) - HOW we read it (operators + trichotomy + tier discipline)

Every Vol 1-10 chapter cites Vol 0 for the ontology. Vol 0 cites itself.

This scaffold specifies the chapter STRUCTURE; Grace's parallel work specifies the chapter CONTENT outline. Combined they form Vol 0 v0.2.

### Proposed chapter structure (10 chapters + 2 appendices)

#### Part I — The Substrate Object

Chapter 1:  $D_{IV}^5$  — the Autogenic Proto-Geometry -  $D_{IV}^5 = SO_0(5,2) / [SO(5) \times SO(2)]$  as Lie group quotient - Bergman geometry primer - Why this geometry (Strong-Uniqueness preview) - Reader exits with: clear mental model of the substrate's mathematical object - Believability anchor: "this is a specific manifold mathematicians have studied since Cartan"

Chapter 2: The Five Integers +  $N_{max}$  - rank = 2,  $N_c = 3$ ,  $n_C = 5$ ,  $C_2 = 6$ ,  $g = 7$ ,  $N_{max} = 137$  - Per-integer "Why X" theorems (T1925, T1930, SP-31-39 consolida-

tions) - The integers as substrate signatures - Reader exits with: concrete handle on the BST primaries - Provability anchor: 6 per-integer theorems registered as D-tier

Chapter 3: The Substrate Operating System - Substrate Working Process Principle (SWPP) — absorption / commitment / emission - Koons tick clock (T2405) - 4-Zone Commitment Cycle (T2420) - 3-Scale Substrate Operation (intra / inter-local / inter-distant) - Reed-Solomon coding on  $GF(2^g)$  (K59) - Reader exits with: substrate as active information-cycle, not static geometry

## Part II — Substrate Symmetries and Structure

Chapter 4: Isotropy Group Structure -  $SO(5) \times SO(2)$  factor decomposition -  $SO(5)$  gives spacetime-side observables -  $SO(2)$  gives internal-symmetry observables (charge, chirality) - Möbius involution gives parity - Full  $SO_0(5,2)$  gives time-translation (Hamiltonian) - Reader exits with: organizing principle for the operator zoo

Chapter 5: Boundary Conditions (the 6+2 BC framework) - Substrate boundary conditions: 6 internal + 2 external - Per-BC theorems (SP-31-40, forthcoming) - Bulk-Boundary Two-Face Structure of integer-webs - Reader exits with: what closes substrate; what couples it to external

Chapter 6: The Integer Web Principle - Each BST primary as a “boundary” holding a web of structural relations - Casey vision Wednesday EOD (filed as Casey-named principle) - Cross-classification matrix (Grace 256-cell) - Reader exits with: integers as anchor-points for substrate’s combinatorial graph

## Part III — Substrate Observables

Chapter 7: The Operator Zoo - Hilbert space specification (SP-31-1, forthcoming Lyra) - Operator inventory: Position (T2419) + Momentum (T2422) + Spin (T2421) + Angular Momentum (#247) + Bell-CHSH (K66, trace-level per Cal #17) + Charge ( $SO(2)$  weight) + Chirality (twistor  $SO(2)$ ) + Number (cycle-count) + Parity (Möbius) + Hamiltonian (pending K52a S24+) + Time (Koons tick) - Organizing principle:  $SO(5) \times SO(2) \times$  Möbius decomposition - Born rule = Bergman projection (K67) - Reader exits with: substrate produces canonical QM observables

Chapter 8: Conservation Laws - Per-conservation-law theorems (SP-31-18, forthcoming Lyra) - Noether on substrate symmetries - Special attention: T (time reversal) and C (charge conjugation) currently NOT EXPLICIT - CPT from  $SO_0(5,2)$  conformal structure - Information conservation via Reed-Solomon coding - Reader exits with: every conservation law derived, not postulated

## Part IV — Substrate Uniqueness and Methodology

Chapter 9: Strong-Uniqueness Theorem - Current state v0.5 (10 criteria, null-model  $(1/3)^{10} \approx 0.002\%$ ) - Per-criterion exposition: C1-C10 + C11-C14 candidates

- Roadmap to v1.0 (curriculum-completion endpoint) - Reader exits with: precise statement of D\_IV<sup>5</sup>'s unique-forcing argument

Chapter 10: Methodology Stack - D/I/C/S tier system - Trichotomy: identifies / derives / predicts - F1-F4 Bridge Object family-member criteria - Mode 6 threshold formalization - K-audit chain governance (Cal/Keeper consensus + Casey Option C) - Believability + Provability dual-axis evaluation - Reader exits with: how BST claims are vetted and tiered

Chapter 11: The Operation (*capstone* — *written 2026-08-16*) - The single act: commitment = WRITE (irreversible binary record) + ENCODE (confined self-validating check); one act, two stages, neither removable - As one operator:  $C = P_{\text{record}} \oplus P_{\text{encode}}$  (the Peirce split); encode = detection/erasure ( $d=2$ ), self-validating NOT unforgeable - The forcing floor: binary commitment  $\mathbb{Z}$  rank 2  $\mathbb{Z}$  isotropy  $\mathbb{Z}$  spin factor  $\mathbb{Z}$  Lorentzian type-IV; the honest four posits (P0 arena, P1 binary, P2a exclusivity, P2b minimality); rank=2 forced EXTERNALLY (first BST integer forced from outside the geometry); dimension the honest residual (Grace's boundary theorem) - Forcing + Evidence =  $2 + 3$  (the information reading); the three readings (QM done / spacetime advanced / SM strong-with-Koide-holdout) - Reader exits with: the one act every observable in the curriculum is a reading of

Chapter 12: Time, Derived (*capstone of the foundations arc* — *written 2026-08-17, Keeper-PASS*) - Time = the flow parameter of the commitment semigroup  $\exp(-\tau J/\hbar)$ ; no clock behind the clock - The generator  $J$  is the linear conformal Hamiltonian (geometry-forced, half-integer spectrum  $E_0=3/2$ ), NOT the Casimir; the arrow = positivity of  $\text{spec}(J)$ ; two faces = the Wick pair (real tick / imaginary circle) - Honesty tier: the degree-2 cover IS spin-statistics (not a novel prediction); charge is internal (degeneracy kills charge=time); gravity's structure is derived, its value one dimensionful input (a theorem — dimensionless content can't make a length) - The reward: BST reduces the dimensionful inputs  $2 \rightarrow 1$  ( $G$  and  $v$  from  $m_e$ ); "trade  $G$  for  $m_e$ " —  $G$  to 0.06% from an atomic mass - Reader exits with: time is derived, and the theory is honest about everything it isn't

## Appendices

Appendix A: Casey-named principles - SWPP (Substrate Working Process Principle) - Five-Absence Predictions Set - Substrate Closure Principle - Graph Forces Principle - Integer Web Principle - Substrate Cognition Network Hypothesis (+ Cosmological Extension, DOUBLE-LOCKED EXTERNAL)

Appendix B: BST  $\leftrightarrow$  Standard Physics Dictionary - Every BST term mapped to standard-physics equivalent (and back) - Example entries: | BST term | Standard term | |—| | substrate | "the vacuum / spacetime structure" | | primary integer | "fundamental constant" | | commitment cycle | "fundamental time unit / Planck cell" | | substrate-CHSH operator | "Bell correlation operator" | | Bergman projection | "Born rule" | | particle-as-winding | "particle as topological excitation" | | Bridge Object | "mathematical structure unifying multiple BST sources" | | Koons

tick | “sub-Planck substrate clock” | - Continuously expanded as Vol 1-10 introduce new terms

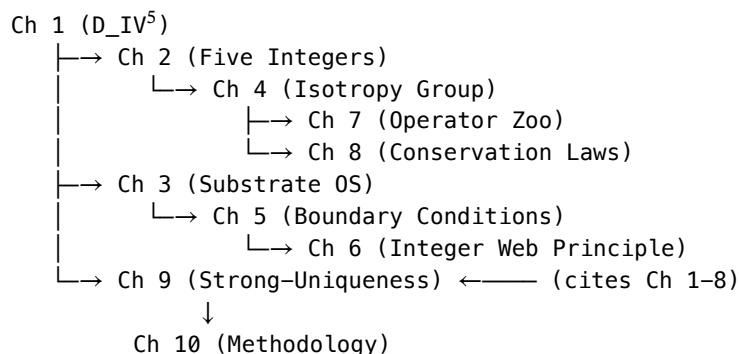
## Theorem-numbering convention for curriculum

Existing: T1-T2427+ (AC graph theorems, sequential across all BST work)

Curriculum chapter cross-references: - Each curriculum chapter cites AC graph theorems with stable T-numbers - New theorems generated during curriculum work get T-numbers via /theorem claim - Vol 0 cites ~80-120 existing theorems + ~10-15 new (from SP-31-39 consolidations + chapter integration work)

Chapter section format: - Vol N, Chapter M, Section K, Subsection L (e.g., “Vol 0 Ch 7 Sec 2.3”) - T-references inline: “via T1925 (Why rank=2)” - AC graph paths shown for derivable claims: “T1 → ... → T1925 → claim”

## Dependency tree (Vol 0 internal)



Reading order matters: Ch 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → 9 → 10. Forward references are minimized; backward references are heavy (Ch 9 cites everything before it).

## Believability + Provability per-chapter rubric

Per Casey directive Wednesday EOD: dual-axis evaluation per chapter. A chapter is DONE only when BOTH axes pass.

Believability axis (passes when): - A physicist reads the chapter and recognizes the content - BST-to-standard-physics dictionary entries exist for all introduced terms - Examples in audience-familiar language (5th-grader test for motivation; PhD-physicist test for technical content) - Cal cold-read verdict: PASS or CONDITIONAL PASS

Provability axis (passes when): - All claims have D/I/C/S tier labels - All D-tier claims have AC graph theorem citations - All C-tier claims have explicit conditional structure - Keeper K-audit verdict: PASS

Status tracking per chapter: | Chapter | Believability | Provability | Overall | |—|—|—  
 |—| | Ch 1 | not started | not started | not started | | Ch 2 | partial (T1925, T1930 done;  
 n\_C, g pending SP-31-39) | partial | partial | | Ch 3 | partial (SWPP + Koons tick + 4-  
 zone filed) | partial | partial | | Ch 4 | partial (operator zoo organizing principle just  
 identified) | partial | partial | | Ch 5 | not started (SP-31-40 pending) | not started  
 | not started | | Ch 6 | partial (Casey-named principle filed; integer-web mapping  
 multi-week) | partial | partial | | Ch 7 | partial (5/11 operators in zoo; rest pending  
 SP-31-6) | partial | partial | | Ch 8 | partial (most laws informal; C/T NOT EXPLICIT)  
 | partial | partial | | Ch 9 | partial (v0.5 filed; v0.6+ pending) | partial | partial | | Ch  
 10 | partial (D/I/C/S + trichotomy + F1-F4 + Mode 6 all filed; M2C2 + Option C  
 documented) | mostly done | mostly done |

## Vol 0 v1.0 completion criteria

- All 10 chapters at Believability PASS + Provability PASS per rubric
- BST↔standard-physics dictionary complete for Vol 0 terms (~50-80 entries)
- All chapter dependencies cross-referenced
- Strong-Uniqueness Theorem at v0.6+ (per Lyra Task #206 extension)
- Cal cold-read of entire volume — PASS
- Keeper K-audit of entire volume — PASS

Estimated cadence: multi-month. Chapters 3, 4, 6, 10 mostly closeable in 4-6 weeks based on existing material. Chapters 5, 7, 8 gated on SP-31 sub-items (SP-31-1 Hilbert space + SP-31-6 operator zoo + SP-31-18 conservation laws + SP-31-40 BCs). Chapter 9 gated on Strong-Uniqueness v0.6+. Chapter 2 gated on SP-31-39 per-integer consolidation (quick).

## Theorem-numbering convention extension proposal

For curriculum-internal theorems (generated DURING curriculum work, not pre-existing in AC graph): - Use prefix CT (Curriculum Theorem) followed by Vol-Ch-Th: e.g., CT0.2.3 = Vol 0 Ch 2 Theorem 3 - CT theorems get T-numbers via /theorem claim and AC graph integration - CT label is a curriculum-internal locator; T-number is the canonical AC graph identifier - Allows chapter-internal theorem-numbering for pedagogy without breaking AC graph hygiene

Casey: review CT proposal. If approved, adopt across all curriculum volumes.

## Companion to Grace's Vol 0 chapter outline (forthcoming Thursday)

This scaffold + Grace's chapter outline = Vol 0 v0.2.

Grace's outline focuses on chapter CONTENT (what each chapter says, what existing material it absorbs). This scaffold focuses on chapter ARCHITECTURE (how chapters relate, theorem numbering, dictionary structure, completion rubric).

When both filed, Lyra absorbs into chapter-draft mode and starts producing chapter drafts. Cal cold-reads as drafts arrive. Keeper audits as drafts pass Cal. Multi-month cadence.

## Status

Curriculum Vol 0 Architectural Scaffold v0.1 FILED Thursday 2026-05-21 morning. Companion to Grace's forthcoming chapter outline. Together = Vol 0 v0.2. 10 chapters + 2 appendices proposed. Theorem-numbering CT convention proposed (Casey review). Believability + Provability per-chapter rubric specified. Vol 0 v1.0 multi-month; gated on SP-31 sub-items + Strong-Uniqueness v0.6+.

— Keeper, 2026-05-21 Thursday 07:45 EDT